

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title

Sudanile Kitchen Mobile Application

Version

1.0

Prepared By

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1. Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for the Sudanile Kitchen Mobile Application, a digital platform designed to preserve, document, and promote South Sudanese culinary heritage through recipes, cooking guides, and cultural food information.

1.2 Scope

The application will provide users with:

- Access to authentic South Sudanese recipes
- Step-by-step cooking instructions
- Recipe search functionality
- Multimedia content (images and videos)
- User accounts and personalization
- Community recipe contributions
- Cultural information related to recipes

The application shall be accessible through Android, iOS, and Web platforms.

2. Overall Description

2.1 Product Perspective

The system is a client-server application consisting of:

- Mobile Frontend
- Web Frontend
- REST API Backend
- PostgreSQL Database
- Cloud Media Storage

2.2 User Classes

Guest User

Can:

- Browse recipes
- Search recipes
- View recipe details

Registered User

Can:

- Save favorites
- Rate recipes
- Submit recipes
- Comment on recipes

Administrator

Can:

- Manage users
 - Approve recipes
 - Manage content
 - View reports and analytics
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3. Functional Requirements

FR1: User Registration

The system shall allow users to register using:

- Email and password
- Google authentication

Inputs

- Name
- Email
- Password

Outputs

- Successful account creation
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FR2: User Authentication

The system shall allow users to:

- Login
 - Logout
 - Reset password
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FR3: Recipe Management

The system shall allow administrators to:

- Create recipes
- Edit recipes
- Delete recipes

Recipe information shall include:

- Title
- Description
- Ingredients
- Cooking steps
- Images
- Videos

- Category
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FR4: Browse Recipes

Users shall be able to:

- Browse recipes
 - Filter by category
 - Sort by popularity
 - Sort by newest
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FR5: Search Functionality

Users shall be able to search recipes using:

- Recipe name
 - Ingredients
 - Categories
 - Keywords
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FR6: Recipe Detail View

The system shall display:

- Recipe title
 - Ingredients list
 - Preparation steps
 - Cooking time
 - Images
 - Videos
 - Cultural information
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FR7: Favorites Management

Users shall be able to:

- Save favorite recipes
 - Remove favorite recipes
 - View favorite recipes
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FR8: Recipe Rating and Reviews

Users shall be able to:

- Rate recipes
 - Write reviews
 - Edit reviews
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FR9: Recipe Submission

Registered users shall be able to submit recipes including:

- Title
- Ingredients
- Instructions
- Images
- Cultural description

Submitted recipes shall require administrator approval.

FR10: Multimedia Management

The system shall support:

- Image uploads
 - Video uploads
 - Video streaming
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FR11: Notifications

The system shall notify users about:

- Recipe approval
 - New recipes
 - Community updates
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FR12: Analytics

Administrators shall view:

- User statistics

- Recipe statistics
 - Search statistics
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4. Non-Functional Requirements

Performance

- Page response time less than 3 seconds.
- Search results displayed within 2 seconds.

Availability

- System uptime of at least 99%.

Security

- Password encryption.
- JWT authentication.
- HTTPS communication.
- Role-based access control.

Scalability

- Support at least 10,000 users initially.
- Support horizontal scaling.

Usability

- Responsive design.
- Intuitive navigation.
- Accessibility support.

Reliability

- Daily backups.
 - Error logging.
 - Disaster recovery procedures.
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5. System Architecture

Frontend

Flutter Mobile Application

Backend

Django REST Framework

Database

PostgreSQL

Storage

Cloudinary

Authentication

JWT + Google OAuth

6. Database Requirements

Users

- user_id
- name
- email
- password_hash
- role

Recipes

- recipe_id
- title
- description
- ingredients
- instructions
- category
- image_url
- video_url

Reviews

- review_id
- user_id
- recipe_id
- rating
- comment

Favorites

- favorite_id
- user_id
- recipe_id

Recipe Submissions

- submission_id
 - user_id
 - status
 - approval_date
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7. Future Enhancements

- AI recipe recommendations
 - Voice-guided cooking
 - Offline recipe access
 - Multilingual support
 - Nutrition analysis
 - Community forums
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8. Acceptance Criteria

The system shall be accepted when:

- Users can register and login successfully.
- Users can browse and search recipes.
- Users can view detailed recipes.
- Users can save favorite recipes.
- Administrators can manage recipes.
- Community recipe submissions function correctly.
- The application performs within defined performance requirements.